

BioCoLoop: COLLABORATIVE AGENTIC RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

Anonymous authors

Paper under double-blind review

ABSTRACT

Biological data are often distributed across institutions, where privacy and data-sharing constraints limit centralized access. Collaborative learning allows multiple laboratories to train a shared model without pooling raw data, but typically fixes the model structure and training hyperparameters in advance, whereas self-improving agents generally assume centralized access to training and evaluation. To address this gap, we introduce BioCoLoop, a collaborative research framework that extends collaboration from parameter fitting to model development itself. BioCoLoop uses an inner loop to train and periodically evaluate proposed model structures or training hyperparameters across laboratories, and an outer loop to use results from completed experiments to generate subsequent proposals. Across drug–target interaction prediction, observed-response proteomic efficacy prediction, and cell perturbation identification, collaborative training across laboratories consistently improves predictive performance. We further show that, with the same total amount of training and evaluation, early evaluation results can improve performance by guiding which models receive additional training. BioCoLoop therefore enables distributed biological data to contribute not only to parameter fitting, but also to the iterative improvement of the model itself.

1. INTRODUCTION

Artificial intelligence is playing an increasingly important role in biological research, from predictive modeling to the generation and evaluation of biological hypotheses (Cui et al., 2024; Theodoris et al., 2023; Roohani et al., 2024). Recent self-improving agents have begun to automate parts of the model-development process (Jiang et al., 2025b; Lu et al., 2024; Zhou et al., 2026). A typical self-improving agent follows an iterative process of proposing, testing, and revising (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2023): it proposes changes to the model structure or training hyperparameters, trains and evaluates the resulting model, and uses the observed results to refine the next proposal, reducing the reliance on manual intervention.

Applying this process to biology, however, is challenging because useful data are often distributed across laboratories and institutions. Different laboratories may study different cell types, perturbations, or experimental assays, and data-sharing constraints may prevent these data from being pooled (Sheller et al., 2020; Bakhtiari et al., 2025; Dayan et al., 2021; Kaissis et al., 2020; Rieke et al., 2020). Collaborative learning addresses part of this problem by allowing multiple laboratories to train a shared model through locally computed model updates (McMahan et al., 2017; Wang et al., 2024; Li et al., 2020; Karimireddy et al., 2020; Kairouz et al., 2021). However, the model structure and training hyperparameters are typically specified before collaborative training begins, which can be restrictive when laboratories hold data with different biological or experimental characteristics.

This limitation also creates an opportunity. Local training and evaluation provide information beyond model updates: they reveal how different model structures or training hyperparameters perform on local data. Although raw data cannot be pooled, these evaluation results can be shared across laboratories and used to guide subsequent experiments. Existing collaborative learning mainly uses locally computed updates for parameter fitting, but rarely uses local evaluation results to determine what should be tested next, whereas existing self-improving agents assume centralized access to both training and evaluation. This leaves a gap between the two settings. *Can the same iterative process*

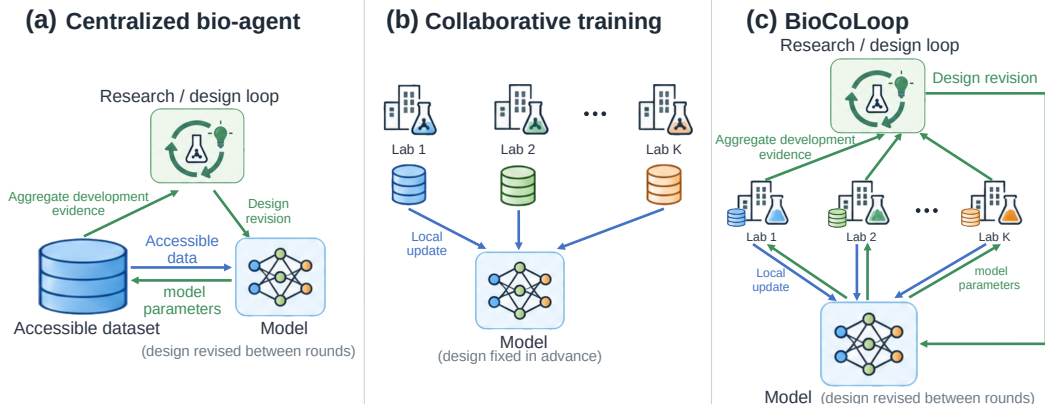


Figure 1: Research settings for biological AI. A centralized bio-agent revises a model using centrally accessible data. Collaborative training allows K laboratories to optimize the same model and training configuration by sharing model updates rather than raw data. BioCoLoop additionally uses results from local evaluation to guide subsequent changes to the model and training configuration.

be carried out across laboratories, so that local training and evaluation jointly guide subsequent model changes without pooling the underlying data?

We introduce BioCoLoop, a collaborative research framework that extends collaboration from parameter fitting to model development itself. Figure 1 contrasts BioCoLoop with centralized self-improving agents and conventional collaborative training. In BioCoLoop, participating laboratories train and evaluate each proposed change on their local data: model updates are aggregated for parameter fitting, and evaluation results are combined to guide what should be tested next.

BioCoLoop consists of an inner collaborative training loop and an outer self-improving loop: the inner loop trains and periodically evaluates proposed model structures or training hyperparameters across laboratories, while the outer loop uses results from completed experiments to generate subsequent proposals and to decide where further training is spent. We evaluate BioCoLoop on drug-target interaction prediction, observed-response proteomic efficacy prediction, and cell perturbation identification, with task-adapted AI-Scientist-v2 and AI-Researcher baselines and both Qwen2.5-7B-Instruct and GPT-5.6 Luna as the self-improving model. Across the evaluated tasks, collaborative training across laboratories consistently improves predictive performance. We also find that early evaluation results can further improve performance by guiding which models receive additional training. Our contributions are summarized as:

- We introduce BioCoLoop, a collaborative research framework that extends collaboration from parameter fitting to model development itself, allowing multiple laboratories to contribute without centralizing their raw data.
- We develop a two-level procedure with an inner loop for collaborative training and evaluation and an outer loop for model development. Evaluation results from the inner loop are used to determine what model structure or training hyperparameters to test next, and which models should receive further training.
- We validate the effectiveness of BioCoLoop on diverse scenarios, including drug-target interaction prediction, observed-response proteomic efficacy prediction, and cell perturbation identification. We further show that, with the same total amount of training and evaluation, using early evaluation results to decide which models continue training can improve performance.

2. RELATED WORK

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed

batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2025).

Self-improving agents in biological research. Self-improving agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist and its successor AI-Scientist-v2 connect ideas to executed experiments, replanning, tree search and writing (Lu et al., 2024; Yamada et al., 2025). DrugEvolve applies research, implementation, evaluation and algorithm memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These systems motivate executable revision and persistent experiment history. Our study asks how such feedback can be supplied by multiple data-owning laboratories through a shared research interface. We instantiate that interface with TAPB for DTL, a ProteinTalks-derived efficacy head for proteomic responses, and scDEBART-based cell representations (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). These models serve as task-specific prediction components; the proposal schema, local training service and evidence-driven selection mechanism remain common across tasks. The strong performance of simple references on some perturbation benchmarks further motivates separating the effect of a research loop from the choice of predictor (Ahlmann-Eltze et al., 2025). Selecting a model specification and its training hyperparameters is also hyperparameter optimization, where random search (Bergstra & Bengio, 2012) and Bayesian optimization (Snoek et al., 2012) are standard references and language models are a recent proposal distribution (Yang et al., 2024); FedEx already couples tuning to collaborative training (Khodak et al., 2021), so direct search is our primary control.

BioCoLoop makes this interface the object of study. A *shared coordinator* resolves each proposed design into a complete executable configuration—a model specification together with its training hyperparameters—fits it across the participating laboratories under one parameter-aggregation rule, returns the resulting *development evidence* to a research model S whose weights θ stay fixed, and appends one evidence card per trial to a persistent evidence set $\mathcal{E}_t$ (Section 3.2). The same interface then drives two evidence-based policies: revising the next proposal from $\mathcal{E}_t$ (Section 3.3) and allocating a fixed training budget from $\mathcal{E}_t$ (Section 3.4). This differs from both lines of work above, which either fix the model specification before collaborative fitting begins or assume centralized evaluation to revise it.

3. METHOD: COLLABORATIVE BIOLOGICAL MODEL IMPROVEMENT

3.1 PROBLEM SETTING AND NOTATION

Consider K laboratories working on the same biological prediction task. Laboratory i holds training data D_i^{train} , with $n_i = |D_i^{\text{train}}|$. Evaluation uses laboratory-local development panels $\{D_i^{\text{dev}}\}_{i \in \mathcal{P}}$, where $\mathcal{P}$ indexes the panels specified by the evaluation protocol. A task specification $\mathcal{T}$ defines the prediction inputs, outputs, training loss and evaluation metric. We write the task predictor as $f_{a,w}$, where $a \in \mathcal{A}$ specifies its design and w its trainable parameters. Given an initial design, a design library $\mathcal{A}$ and a research budget, BioCoLoop seeks a design and fitted predictor that perform well on the designated evaluation population. The three task families instantiate $f_{a,w}$ as task-specific predictors within this shared research procedure.

Our design library varies four choices: learning rate, weight decay, coordinator momentum and an optional residual prediction module. A `design_id` names one configuration of these choices. Thus, in the reported study a candidate is an executable model specification together with its training hyperparameters, not an arbitrary isolated neural block: MLP or MHA alternatives would be candidates only if they were explicitly registered in the design library. Here the only architecture toggle in the menu is a zero-initialized, width-32 residual prediction module; the task-specific backbone and prediction head are supplied by the adapter.

Here *coordinator* means the trusted execution/orchestration service, not the self-improving model, a trainable neural layer, or an additional laboratory. It resolves a registered design identifier, initializes and broadcasts the predictor, coordinates local fitting, aggregates local model states, aggregates

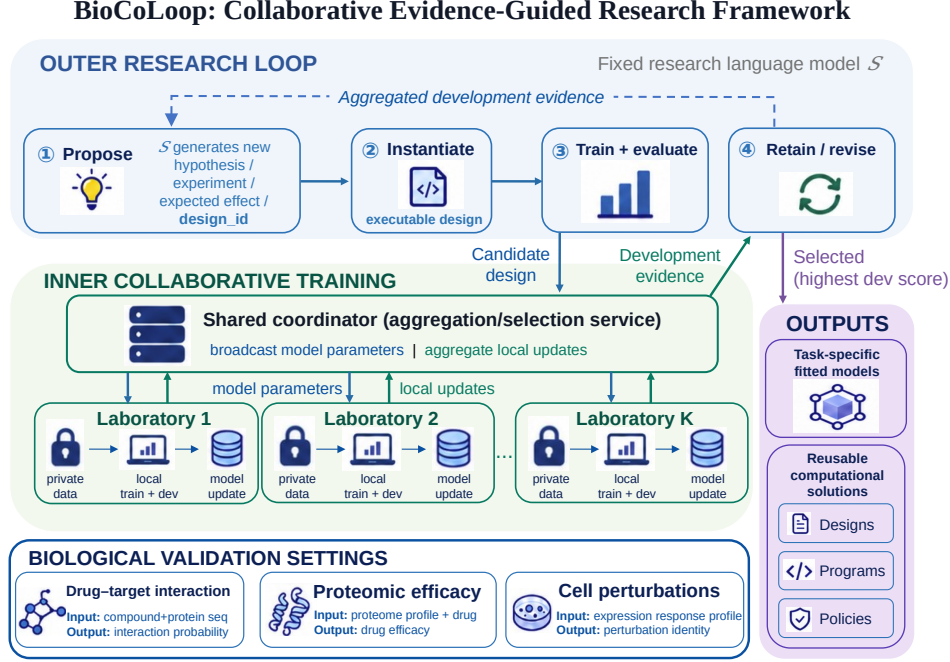


Figure 2: Data flow in BioCoLoop and the symbols used throughout. The outer research loop is driven by the research model $\mathcal{S}$ with fixed weights θ , which proposes a candidate model specification and training hyperparameters and receives the aggregated development evidence $\mathcal{E}_t$ after each evaluation. The shared coordinator of Section 3.1 is the logical service that resolves the candidate, distributes model parameters to K laboratories, fits them locally, aggregates the returned updates and summarizes the development evidence. The next proposal is conditioned on $\mathcal{E}_t$ through Equation 3. Task adapters implement drug-target interaction prediction, observed-response proteomic efficacy prediction and cell perturbation identification.

bounded development measurements, and applies the checkpoint/design-selection rule. *Shared coordinator* means that all participating laboratories use this same logical protocol and decision point: they receive the same candidate state and schedule, while their raw training and development measurements remain local. The coordinator returns only the parameter state or update information needed for collaborative fitting and the declared aggregate scores/diagnostics needed for research decisions. The harness therefore coordinates candidate selection, local fitting, evaluation and the return of evidence to the research model $\mathcal{S}$ (Figure 2), a language model whose weights θ stay fixed throughout a run. We use the common fitting interface to study two policies: history-guided proposal revision (Section 3.3) and trajectory-guided training allocation (Section 3.4). The experiments simulate separate laboratories on one host.

3.2 COLLABORATIVE FITTING AND SHARED EVALUATION

For each candidate a , the coordinator resets the predictor to its prescribed initialization w_0 and distributes it to all training laboratories. Let LocalEpoch_a denote one full local AdamW epoch with the candidate’s learning rate and weight decay, batch size 64 and gradient-norm clipping at one. The optimizer state resets at each round. With sample weights $\pi_i = n_i / \sum_{j=1}^K n_j$, coordinator momentum coefficient β_a and initial velocity $v_0 = 0$, round r updates the trainable parameters by

$$u_{i,r} = \text{LocalEpoch}_a(w_{r-1}; D_i^{\text{train}}), \quad \bar{w}_r = \sum_{i=1}^K \pi_i u_{i,r}, \quad (1)$$

$$v_r = \beta_a v_{r-1} + \bar{w}_r - w_{r-1}, \quad w_r = w_{r-1} + v_r.$$

The coordinator carries momentum across aggregation rounds through v_r ; $\beta_a = 0$ gives sample-weighted parameter averaging (McMahan et al., 2017). In one round, the coordinator sends the

current trainable state and fixed candidate settings to each laboratory; laboratory i returns its locally fitted state $u_{i,r}$ (and its training-example count n_i) for aggregation. Frozen feature extractors remain unchanged. Task adapters supply the biological inputs, trainable prediction components and local loss. The laboratories never send raw examples, expression profiles, labels or per-example identifiers to the coordinator or self-improving agent.

Every five rounds and at the final round, the laboratories evaluate the aggregated predictor f_{a,w_r} on the development panels indexed by $\mathcal{P}$. Each laboratory returns scalar primary score $s_i(a, r)$, development loss $\ell_i^{\text{dev}}(a, r)$ and the bounded trajectory diagnostics required by the protocol; it does not return its underlying observations. In the main comparison these are the participating laboratories' panels; fixed-pool repartitioning retains all ten original panels, and source-transfer studies use the original target panels. Define $s_i(a, r)$ and $\ell_i^{\text{dev}}(a, r)$ as the primary score and loss on panel i . The coordinator computes

$$\bar{s}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} s_i(a, r), \quad \bar{\ell}(a, r) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \ell_i^{\text{dev}}(a, r). \quad (2)$$

The selected round $\hat{r}_a$ maximizes $\bar{s}(a, r)$, with lower $\bar{\ell}(a, r)$ breaking ties and the earlier round retained on a complete tie. Its parameters are $\hat{w}_a = w_{\hat{r}_a}$; the pair $q(a) = (\bar{s}(a, \hat{r}_a), -\bar{\ell}(a, \hat{r}_a))$ ranks the design. In plain terms, selection takes the highest equal-laboratory mean development score, then the lower mean development loss, then the earlier checkpoint. Training therefore weights examples, whereas selection weights the designated development panels equally. The main proposal-search experiments assign each candidate 100 training rounds; training duration is specified by the evaluation schedule, not by `design_id`. Each candidate therefore produces one *development-evidence card*, and after t trials the accumulated *development evidence* is the ordered set $\mathcal{E}_t = \{e_1, \dots, e_t\}$. Card e_j records the executed configuration a_j , i.e. its model specification and training hyperparameters; the hypothesis h_j ; the best development primary score and loss (s_j, ℓ_j) with the selected round $\hat{r}_j$; the endpoint diagnostics Δ_j , namely the first and last development-evaluation rounds with their sample-weighted training losses, the equal-laboratory development scores and losses at those rounds and the changes between them; the optional worst-laboratory score and evaluating-laboratory count; and the acceptance decision dec_j . The card is therefore more than a single metric, but it is not raw data or free-form biological evidence.

For auxiliary-source studies with different prediction targets, shared encoders follow the same update rule, while each source-specific head is aggregated only over laboratories using that prediction target. Checkpoint selection then uses the original target's development panels. This allows auxiliary measurements to improve the shared representation while preserving each source's prediction objective.

3.3 PROPOSAL REVISION FROM DEVELOPMENT EVIDENCE

The first proposal is generated once without development feedback and shared by direct and loop search. History feedback begins with the second proposal. For subsequent research steps t , let a_t be the retained design, $\mathcal{E}_t$ the development evidence accumulated so far and $\mathcal{U}_t$ the untried configurations. The research model S , with weights θ held fixed, generates a proposal p_t and the coordinator maps its `design_id`, denoted $\text{id}(p_t)$, to the candidate:

$$p_t \sim P_\theta(\cdot \mid \mathcal{T}, a_t, \mathcal{E}_t, \mathcal{U}_t), \quad \tilde{a}_t = \text{Compile}(\text{id}(p_t)). \quad (3)$$

The proposal also contains a hypothesis, an experiment and an `expected_effect`: these describe the motivation, intended comparison and anticipated outcome. They provide an interpretable account of the proposed experiment. The identifier determines the executed configuration, and measured development performance determines selection; the prose cannot override the registered configuration or the measured result.

After training $\tilde{a}_t$, the coordinator appends the card e_t of Section 3.2 to the development evidence. Using lexicographic comparison of q —higher score, then lower loss—the research state becomes

$$a_{t+1} = \begin{cases} \tilde{a}_t, & q(\tilde{a}_t) >_{\text{lex}} q(a_t), \\ a_t, & \text{otherwise,} \end{cases} \quad \mathcal{E}_{t+1} = \mathcal{E}_t \parallel [e_t], \quad (4)$$

where $\parallel$ appends a card to the evidence set. Unsuccessful trials remain in that set; invalid proposals and numerically failed fits consume a proposal slot and leave the incumbent unchanged. Subsequent

proposals therefore respond to measured outcomes through updated context, without fine-tuning the self-improving model. After the proposal budget is exhausted, the retained design and its selected checkpoint are evaluated on the held-out task.

3.4 EVIDENCE-GUIDED TRAINING ALLOCATION

We use trajectory-guided allocation in a controlled study (Section 4.4) to test whether early evaluation evidence improves the use of a fixed training budget. The screening–promotion policy retains both high-scoring and still-improving candidates. It screens ten predetermined designs for 20 rounds, evaluating every five rounds through Equation 2. Within each of three learning-rate groups, it promotes the best-scoring design and, among the remaining designs, the one with the largest score improvement from round 5 to round 20. Development loss breaks score ties. The six promoted designs restart from their initial parameters for 100-round training; their best development checkpoints determine the final predictor.

The control trains all ten designs for 80 rounds. Both schedules use 800 aggregation rounds and 160 aggregate development evaluations; the scheduler does not generate language-model proposals. This separate allocation study improves Norman Top-1 by 10.84 percentage points on average (Section 4.4). The main comparison and proposal-history experiments use 100 training rounds per candidate to compare design selection at a fixed training duration; they do not use the allocation policy. Appendix A collects the complete proposal schema, algorithm and allocation rules.

4. EXPERIMENTS

4.1 TASKS, DATASETS AND EVALUATION

We evaluate BioCoLoop on three biological prediction problems. Each task uses its own base model, while collaborative training, evaluation, and proposal generation follow the same procedure. We consider two ways of forming laboratories. In the main setting, each dataset is divided into ten disjoint groups. In a complementary setting, distinct biological datasets or experimental scenarios are treated as separate laboratories. The first setting provides a controlled comparison of collaborative training, while the second tests whether information can transfer across genuinely different data sources.

Drug-target interactions. Given a compound and a protein sequence, the model predicts whether they interact. BindingDB provides 50,149 training pairs and 5,604 development pairs, with canonical compounds defining laboratory assignments. We use TAPB as the base model, with frozen ESM2 protein features and trainable drug, fusion, and classification modules (Lin et al., 2025). The primary metric is AUROC on 5,505 random test pairs. Additional test sets containing 3,791 unseen-drug pairs and 2,507 unseen-protein pairs measure generalization to unseen entities. AP, MCC, accuracy, and cross-entropy are reported as secondary metrics.

Observed-response proteomics. Given a drug representation and its measured 6h and 24h proteomic responses, the model predicts binary efficacy. A ProteinTalks PTPC subset provides 386 training, 105 development, and 148 test conditions (Sun et al., 2026). Compound-group assignment gives each laboratory 49-50 training and development conditions. The base model is a ProteinTalks-derived efficacy model with protein and drug branches, trained from random initialization. Average precision (AP) is the primary metric. AUROC, accuracy, and cross-entropy are secondary metrics. Unlike de novo response prediction, this task conditions on observed post-treatment proteomic measurements.

Cell perturbation identification. Given an expression response and five candidate interventions, the model identifies the intervention whose predicted response has the highest cosine similarity to the observation. We use VCC single-gene perturbations (Arc Institute, 2025), Norman double-gene perturbations (Norman et al., 2019), and a Tahoe subset of drug perturbations in A549 at 5 μ M (Zhang et al., 2025). Intervention-disjoint training, development, and test counts are 60/20/20, 70/20/27, and 180/60/67, respectively. The base model uses a frozen scDEBART backbone with a trainable response head (Sung & Kim, 2026). All methods use the same corrected attention masks and the same candidate sets. The primary metric is intervention-macro Top-1, which first averages accuracy within each intervention and then across interventions. Random selection has an expected Top-1 of 20%. Micro Top-1, MRR, Top-3, and cross-entropy are reported as secondary metrics. Appendix A

Table 1: Main comparison by self-improving model (mean $\pm$ sample SD, seeds 42–44; scores $\times 100$). Public self-improving agents use laboratory 0 and BioCoLoop ten under a shared design library, fits and held-out scorer. Bold marks the best complete value per endpoint and the lowest mean rank; underline marks the runner-up; $\dagger$ partial and – unscored cells are not imputed. Mean rank ranks the methods of a block from 1 (best) to 5 on every endpoint they completed and averages those ranks, so lower is better and the $\uparrow/\downarrow$ offset is the gap to that block’s fixed reference. The three cell endpoints are five-option identification tasks with a 20% chance level, so their headroom above chance is small. $\star$ marks a cell whose controller was admitted under an extended native-context budget; the cell it does not score is GPT-5.6 Luna AI-Researcher DTI, whose native trajectory did not finish inside that budget (Appendix C).

Model / method	DTI	Proteomics	Cell perturbation			Average
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$	Mean rank $\downarrow$
Qwen2.5-7B-Instruct						
Fixed model	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	2.40
Direct	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21	3.40 $\downarrow$ 1.0
AI-Scientist-v2	89.68 $\pm$ 0.80	20.18 $\pm$ 0.43	18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	15.42 $\pm$ 6.21	3.80 $\downarrow$ 1.4
AI-Researcher	86.09 $\pm$ 0.88	20.12 $\pm$ 0.05	19.70 $\pm$ 2.87 $\dagger$	21.73 $\pm$ 6.63	17.41 $\pm$ 6.21	3.75 $\downarrow$ 1.4
BioCoLoop	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59	1.20 $\uparrow$ 1.2
GPT-5.6 Luna (low reasoning)						
Fixed model	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	2.20
Direct	89.89 $\pm$ 1.02	20.17 $\pm$ 0.42	18.61 $\pm$ 2.78	21.75 $\pm$ 6.59	15.92 $\pm$ 6.03	3.20 $\downarrow$ 1.0
AI-Scientist-v2	89.59 $\pm$ 0.95	23.37 $\pm$ 5.89	18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	11.94 $\pm$ 2.11 $\dagger$	3.25 $\downarrow$ 1.0
AI-Researcher	–	20.29 $\pm$ 0.27 $\star$	18.61 $\pm$ 2.78 $\star$	14.12 $\pm$ 0.04 $\star$	15.42 $\pm$ 6.21 $\star$	3.75 $\downarrow$ 1.5
BioCoLoop	93.67 $\pm$ 0.98	37.11 $\pm$ 0.01	31.50 $\pm$ 0.00	<u>22.69</u> $\pm$ 0.73	22.39 $\pm$ 2.59	1.20 $\uparrow$ 1.0

provides the response construction, chemical conditioning, training losses, and secondary evaluation details.

4.2 COMPARED METHODS AND SHARED PROTOCOL

We compare BioCoLoop with three types of baselines. The *fixed model* trains the task-specific base model once using a predefined model structure and training hyperparameters, without proposal generation or iterative model improvement. *Direct optimization* uses the same proposal generator and the same allowed model structures and training hyperparameters as BioCoLoop, but generates each proposal without access to results from previous experiments. It therefore tests whether searching over model and training choices is useful without iterative feedback. AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) are adapted to the same biological tasks and use the same training and evaluation interface.

BioCoLoop additionally provides results from completed experiments when generating subsequent proposals and performs collaborative training and evaluation across laboratories. We repeat the comparison using Qwen2.5-7B-Instruct and GPT-5.6 Luna as proposal generators (OpenAI, 2026). All methods share the same task-specific base model, allowed model structures and training hyperparameters, and held-out evaluation. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory, while BioCoLoop is reported at one and ten laboratories; the control comparison in Appendix A (Table A.1) therefore holds the research loop fixed and varies only laboratory participation. Collaborative learning exists precisely to use data that institutions will not or cannot pool (McMahan et al., 2017; Sheller et al., 2020; Kaissis et al., 2020), and to our knowledge BioCoLoop is the first framework to pair collaborative fitting with a self-improving research loop. Full implementation and training details are provided in Appendices A and C.

4.3 COLLABORATIVE ACCESS IMPROVES TASK PERFORMANCE

Table 1 presents the Qwen2.5-7B-Instruct and GPT-5.6 Luna blocks over seeds 42–44. The blocks are not matched in access: the fixed and direct references and the public self-improving agents use laboratory 0, while BioCoLoop uses the ten-laboratory pool that collaborative learning is designed to exploit, so we read the table against the matched-access control of Appendix A (Table A.1). Cells that terminate before a valid held-out selection remain marked rather than imputed (Appendix C).

Holding the search mode fixed, increasing participation from one to ten laboratories adds +3.87 AUROC points for DTI, +16.85 AP points for proteomics, and +11.32, +8.10 and +6.97 Top-1 points

Table 2: Evidence-guided training allocation at a matched budget. Held-out scores use the same ten designs, ten laboratories, 800 full-client aggregation rounds and 160 aggregate development evaluations per method. Uniform allocation gives every design 80 rounds; the evidence-guided scheduler screens all designs for 20 rounds and restarts six promoted designs for 100-round training. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [-1.00, 7.46]. The separate DTI development replay is in Appendix A.

Benchmark	Uniform	Evidence-guided	Gain	W/T/L
Proteomics: efficacy	37.47	37.47	+0.00	0/3/0
Cell: VCC single-gene	20.01	20.01	+0.00	0/3/0
Cell: Norman double-gene	11.48	22.32	+10.84	3/0/0
Cell: Tahoe drug	20.90	22.89	+1.99	1/2/0

for VCC, Norman and Tahoe, so collaborative access improves four of the five endpoints, with the largest gains on the drug-target and proteomic tasks. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory only; the ten-laboratory row of Table A.1 is the collaborative configuration reached with measured feedback. The identification endpoints have the least headroom—chance is 20% and the best displayed Norman value is 25.56—so we treat them as a low-headroom stress test and base the primary conclusions on the drug-target and proteomic endpoints. Appendix A.6 gives the chance-level and representation analysis behind these values.

4.4 MEASURED TRAJECTORIES IMPROVE TRAINING ALLOCATION

We apply the policy in Section 3.4 to the same ten designs and ten laboratories under both schedules; each spends 800 aggregation rounds and 160 aggregate development evaluations (Table 2). Trajectory-guided allocation moves Norman from 11.48% to 22.32% Top-1: a +10.84-point mean effect, positive in every seed, with hierarchical paired-bootstrap 95% CI [4.54, 17.73]. Chance is 20% on this five-option task, so the uniform schedule sits below chance and the evidence-guided schedule recovers above it, restoring usable signal rather than adding a large margin. Across all twelve task-seed comparisons (seeds 53–55) the policy gives four wins and eight ties; Tahoe gains 1.99 points on average, while VCC and proteomics tie. The VCC level here is not comparable with Table 1 (31.54): this study uses its own menu and seeds 53–55 (Appendix B). The feedback channel’s measurable held-out effect is therefore confined to allocation.

4.5 LEARNING FROM DIFFERENT LABORATORY SCENARIOS

Source-as-laboratory experiments replace random shards with distinct biological collections: BindingDB, BioSNAP, Davis and Human for DTI, three external single-cell studies for VCC, and the target PTPC collection plus Lin, Ruprecht and decryptE for proteomics (Lin et al., 2026; Ruprecht et al., 2020; Eckert et al., 2025). The target’s development panels select checkpoints and a common target test measures transfer (Appendix B gives identities, sizes and exclusions). External measurements can improve target prediction: adding all three proteomic studies raises AP from 34.41 to 37.53, and Davis and Human raise DTI AUROC from 85.47 to 87.86 and 87.95. Source compatibility nevertheless matters alongside data volume, and at matched participation and condition count heterogeneous updates do not improve VCC identification (28.82% versus 30.70%).

4.6 HOW MANY LABORATORIES SHOULD PARTICIPATE?

We vary $K \in \{1, 2, 5, 10\}$ in two ways: adding the first K laboratories increases available data, while merging the full pool into K groups holds total data fixed. Both use the default design and 100 rounds (DTI seed 61, other endpoints seeds 61–63).

Figure 3 separates access from partitioning. Increasing participation raises DTI AUROC from 83.65 to 93.88; with the DTI pool fixed, $K = 1, 2, 5, 10$ gives 85.47, 88.96, 88.05 and 93.88. The fixed-pool optimum is task specific: proteomics peaks at 39.19 AP for $K = 5$, whereas Norman falls from 49.63 to 15.48 Top-1 between $K = 1$ and $K = 10$, crossing below chance.

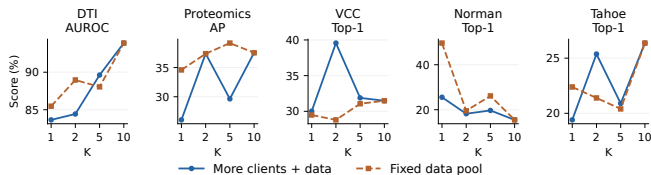


Figure 3: Laboratory-count sensitivity. Solid lines add participating laboratories and data; dashed lines repartition the full training pool. Points are means over the study seeds; full values and SD are in Appendix B. Each panel uses its task’s primary metric; tested K settings are equally spaced.

4.7 HOW DOES FEEDBACK CHANGE THE RESEARCH TRAJECTORY?

Direct and loop share the designs, initial proposal, training schedule and selection rule; only loop receives measured history (Qwen2.5-7B-Instruct and gpt-5.6-luna, all five endpoints, DTI seed 42 and other tasks seed 61). Appendix B reports development-selected and held-out test performance at each prespecified proposal prefix.

Measured evidence changes how quickly the search reaches a good design. Over the 36-design, 24-slot library the model reaches the common development target at slot 4 with history against slot 22 without it, and across ten short direct/loop pairs Norman/Luna improves by 0.81 Top-1 points while eight pairs tie.

5. CONCLUSION

BioCoLoop turns laboratory-local training and evaluation into two shared resources: aggregated model updates for collaborative fitting and aggregated development evidence for deciding what to propose and where to spend training. Collaborative access is the dominant effect across the studied endpoints, with the largest gains in drug–target and proteomic prediction, while the cell identification tasks remain a low-headroom stress test near chance. Design search and evidence-guided allocation each improve the endpoint whose candidate set has measurable headroom, and independent sites, larger libraries and a random-sweep control are the natural next steps (Appendix A.6).

REPRODUCIBILITY STATEMENT

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, data partitions, reference models and metrics. Appendix A provides the executable proposal schema, complete design library, optimization settings and matched-budget promotion rule. Appendix B specifies the laboratory-count, source-transfer and proposer-budget studies. The accompanying Statistical Supplement gives complete per-seed contrasts and confidence intervals for every reported contrast, including non-positive differences. An anonymized release accompanies the submission and contains the implementation, task adapters, experiment configurations, the full proposal log and the prediction-level evaluation records, bound to code and data hashes, random seeds, development trajectories, selection decisions and saved predictions. The proposal log records all 504 attempted slots (20×6 for the six-slot protocol plus 16×24 for the extended protocol) with their configuration changes; the appendix presents representative decisions and complete performance summaries. Aggregate tables are generated from these completed records, with independent score recomputation for the main comparison and completed follow-up evaluations. Two limits are stated explicitly: the local Qwen block is seed-reproducible, while the Luna service exposes no generation seed and is not bit-reproducible, and the public self-improving agents are task-adapted re-implementations whose disabled native components are listed in Appendix C.

ETHICS STATEMENT

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. The experiments simulate separate laboratories on one host. Retaining raw data at each laboratory avoids direct transfer to the self-improving model; parameter updates and aggregate diagnostics would still require an explicit privacy and security policy in a multi-institutional deployment. Predictions are evaluated as research outputs, not as validated biological or therapeutic interventions.

AI USE STATEMENT

The main comparison retains matched proposal records for local Qwen2.5-7B-Instruct and the service identified as gpt-5.6-luna. Luna is run at low reasoning effort through a text-only bridge with tools disabled; its service does not expose a generation seed, so the seed labels identify the training/search repetitions rather than deterministic language-model sampling. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, and manuscript drafting and revision. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

REFERENCES

- Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8): 1657–1661, 2025. doi: 10.1038/s41592-025-02772-6. URL <https://doi.org/10.1038/s41592-025-02772-6>.
- Arc Institute. Virtual cell challenge. Arc Virtual Cell Atlas dataset, 2025. URL <https://github.com/ArcInstitute/arc-virtual-cell-atlas/tree/4f7bb170c61d2fafce70f67db6e52c7d9af1be4b/virtual-cell-challenge>. 2025 data release; training file adata_Training.h5ad. Documentation accessed September 22, 2026.
- Mohammad Bakhtiari, Stefan Bonn, Fabian Theis, Olga Zolotareva, and Jan Baumbach. FedscGen: privacy-preserving federated batch effect correction of single-cell RNA sequencing data. *Genome Biology*, 26:216, 2025. doi: 10.1186/s13059-025-03684-6. URL <https://doi.org/10.1186/s13059-025-03684-6>.

- James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. In *Journal of Machine Learning Research*, volume 13, pp. 281–305, 2012. URL <https://jmlr.org/papers/v13/bergstra12a.html>.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0. URL <https://doi.org/10.1038/s41592-024-02201-0>.
- Ittai Dayan, Holger R. Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z. Abidin, Andrew Liu, Anthony Beardsworth Costa, et al. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine*, 27(10):1735–1743, 2021. doi: 10.1038/s41591-021-01506-3. URL <https://doi.org/10.1038/s41591-021-01506-3>.
- Jiayuan Ding, Jianhui Lin, Ziyang Miao, Nils Mechtel, Shiyu Jiang, Yixin Wang, Zhaoyu Fang, Jorge D. Martin-Rufino, Chen Weng, Reuben Saunders, Weize Xu, Jonathan S. Weissman, Min Li, Jiliang Tang, Wei Ouyang, Yuancheng Ryan Lu, and Xiaojie Qiu. Predictive single cell foundation model for gene regulation and aging with privacy-preserving tabular learning, 2026. URL <https://arxiv.org/abs/2607.19400v1>. Version 1.
- Stephan Eckert, Nicola Berner, Karl Kramer, Annika Schneider, Julian Müller, Severin Lechner, Sarah Brajkovic, Amirhossein Sakhteman, Christian Graetz, Jonas Fackler, Michael Dudek, Michael W. Pfaffl, Percy Knolle, Stephanie Wilhelm, and Bernhard Kuster. Decrypting the molecular basis of cellular drug phenotypes by dose-resolved expression proteomics. *Nature Biotechnology*, 43:406–415, 2025. doi: 10.1038/s41587-024-02218-y. URL <https://doi.org/10.1038/s41587-024-02218-y>. First published online 7 May 2024.
- GAIR-NLP DrugEvolve contributors. DrugEvolve: Official source and applications, 2026. URL <https://github.com/GAIR-NLP/DrugEvolve/tree/e945c7d0da28ddce73904419116a51e03921e6e6>. Commit e945c7d0da28ddce73904419116a51e03921e6e6.
- HKUDS. AI-Researcher: Official source, 2025. URL <https://github.com/HKUDS/AI-Researcher/tree/f9a6f8480860c193afff600eeffe3defcee8a978>. Commit f9a6f8480860c193afff600eeffe3defcee8a978.
- Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. MLAGentBench: Evaluating language agents on machine learning experimentation, 2023. URL <https://arxiv.org/abs/2310.03302v1>. Version 1, 5 October 2023.
- Longda Jiang, Carol Dalgarno, Efthymia Papalexi, Isabella Mascio, Hans-Hermann Wessels, Huiyoun Yun, Nika Iremadze, Gila Lithwick-Yanai, Doron Lipson, and Rahul Satija. Systematic reconstruction of molecular pathway signatures using scalable single-cell perturbation screens. *Nature Cell Biology*, 27:505–517, 2025a. doi: 10.1038/s41556-025-01622-z. URL <https://doi.org/10.1038/s41556-025-01622-z>.
- Zhengyao Jiang, Dominik Schmidt, Dhruv Srikanth, Dixing Xu, Ian Kaplan, Deniss Jacenko, and Yuxiang Wu. AIDE: AI-driven exploration in the space of code, 2025b. URL <https://arxiv.org/abs/2502.13138v1>. Version 1, 18 February 2025.
- Peter Kairouz, H. Brendan McMahan, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021. doi: 10.1561/22000000083. URL <https://doi.org/10.1561/22000000083>.
- Georgios A. Kaissis, Marcus R. Makowski, Daniel Rückert, and Rickmer F. Braren. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, 2(6):305–311, 2020. doi: 10.1038/s42256-020-0186-1. URL <https://doi.org/10.1038/s42256-020-0186-1>.
- Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J. Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning, 2020. URL <https://arxiv.org/abs/1910.06378v1>. Version 1, 14 October 2019; published at ICML 2020.

- Mikhail Khodak, Renbo Tu, Tian Li, Liam Li, Maria-Florina Balcan, Virginia Smith, and Ameet Talwalkar. Federated hyperparameter tuning: Challenges, baselines, and connections to weight-sharing. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://papers.neurips.cc/paper_files/paper/2021/hash/a0205b87490c847182672e8d371e9948-Abstract.html.
- Haoyuan Li, Mathias Funk, and Aaqib Saeed. Helmsman: Autonomous synthesis of federated learning systems via collaborative LLM agents, 2025. URL <https://arxiv.org/abs/2510.14512v2>. Version 2, 19 December 2025.
- Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks, 2020. URL <https://arxiv.org/abs/1812.06127v1>. Version 1, 14 December 2018; published at MLSys 2020.
- Chuwei Lin, Catherine M. Snizek, Christopher D. McGann, Rashmi Karki, Ross M. Giglio, Benjamin A. Garcia, José L. McFaline-Figueroa, and Devin K. Schweppe. Defining the heterogeneous molecular landscape of lung cancer cell responses to epigenetic inhibition. *Communications Biology*, 9:176, 2026. doi: 10.1038/s42003-025-09455-0. URL <https://doi.org/10.1038/s42003-025-09455-0>. Published 28 January 2026.
- Gaoming Lin, Xin Zhang, Zhonghao Ren, Quan Zou, Prayag Tiwari, Changjun Zhou, and Yijie Ding. TAPB: an interventional debiasing framework for alleviating target prior bias in drug-target interaction prediction. *Nature Communications*, 16:10867, 2025. doi: 10.1038/s41467-025-66915-1. URL <https://doi.org/10.1038/s41467-025-66915-1>.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery, 2024. URL <https://arxiv.org/abs/2408.06292v1>. Version 1, 12 August 2024.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative refinement with self-feedback, 2023. URL <https://arxiv.org/abs/2303.17651v1>. Version 1, 30 March 2023.
- Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pp. 1273–1282, 2017. URL <https://proceedings.mlr.press/v54/mcmahan17a.html>.
- Nils Mechtel, Wei Ouyang, and Chiron Platform contributors. Chiron Platform: Federated single-cell training and agent interfaces, 2026. URL <https://github.com/aristoteleo/chiron/tree/3241b35c8b516c126aa68da450708de3da717ee0>. Commit 3241b35c8b516c126aa68da450708de3da717ee0.
- Ajay Nadig, Joseph M. Replogle, Angela N. Pogson, Mukundh Murthy, Steven A. McCarroll, Jonathan S. Weissman, Elise B. Robinson, and Luke J. O’Connor. Transcriptome-wide analysis of differential expression in perturbation atlases. *Nature Genetics*, 57:1228–1237, 2025. doi: 10.1038/s41588-025-02169-3. URL <https://doi.org/10.1038/s41588-025-02169-3>.
- Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019. doi: 10.1126/science.aax4438. URL <https://doi.org/10.1126/science.aax4438>.
- OpenAI. The GPT-5.6 model family: the gpt-5.6-luna service, 2026. Proprietary reasoning service accessed through the OpenAI Codex CLI at low reasoning effort with tools disabled; the service exposes no generation seed. Accessed September 2026.
- Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, Jeffrey A. Hussmann, Alexander Lenail, Alina Guna, Lauren Mascibroda, Eric J. Wagner, Karen Adelman, Gila Lithwick-Yanai,

- Nika Iremadze, Florian Oberstrass, Doron Lipson, Jessica L. Bonnar, Marco Jost, Thomas M. Norman, and Jonathan S. Weissman. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013. URL <https://doi.org/10.1016/j.cell.2022.05.013>.
- Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletari, Holger R. Roth, Shadi Albarqouni, Spyridon Bakas, Mathieu N. Galtier, Bennett A. Landman, Klaus Maier-Hein, Sébastien Ourselin, Micah Sheller, Ronald M. Summers, Andrew Trask, Daguang Xu, Maximilian Baust, and M. Jorge Cardoso. The future of digital health with federated learning. *npj Digital Medicine*, 3:119, 2020. doi: 10.1038/s41746-020-00323-1. URL <https://doi.org/10.1038/s41746-020-00323-1>.
- Yusuf Roohani, Andrew Lee, Qian Huang, Jian Vora, Zachary Steinhart, Kexin Huang, Alexander Marson, Percy Liang, and Jure Leskovec. BioDiscoveryAgent: An AI agent for designing genetic perturbation experiments, 2024. URL <https://arxiv.org/abs/2405.17631v1>. Version 1, 27 May 2024.
- Benjamin Ruprecht, Julie Di Bernardo, Zhao Wang, Xuan Mo, Oleg Ursu, Matthew Christopher, Rafael B. Fernandez, Li Zheng, Brian D. Dill, Huijun Wang, Yuting Xu, Andy Liaw, Jonathan D. Mortison, Nirodhini Siriwardana, Brian Andresen, Meir Glick, James R. Tata, Victoria Kutilek, Ivan Cornella-Taracido, and An Chi. A mass spectrometry-based proteome map of drug action in lung cancer cell lines. *Nature Chemical Biology*, 16:1111–1119, 2020. doi: 10.1038/s41589-020-0572-3. URL <https://doi.org/10.1038/s41589-020-0572-3>.
- SakanaAI. The AI scientist-v2: Official source, 2025. URL <https://github.com/SakanaAI/AI-Scientist-v2/tree/96bd51617cfd5b494a9fc283af00fe090edfae48>. Commit 96bd51617cfd5b494a9fc283af00fe090edfae48.
- Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10:12598, 2020. doi: 10.1038/s41598-020-69250-1. URL <https://doi.org/10.1038/s41598-020-69250-1>.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning, 2023. URL <https://arxiv.org/abs/2303.11366v1>. Version 1, 20 March 2023.
- Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. Practical bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, 2012. URL <https://arxiv.org/abs/1206.2944>.
- Rui Sun, Liujia Qian, Yongge Li, Tong Liu, Honghan Cheng, Xuedong Zhang, Xueya Zhou, Yuecheng Zhan, Guangmei Zhang, Zhengchao Luo, Kunpeng Ma, Chunlong Wu, Dongchen Ji, Zhangzhi Xue, Hongxue Meng, Yuhang Xiang, Dingwei Lei, Qianhe Zhou, Wenbin Hu, Yuhang Deng, Lingling Tan, Qi Xiao, Zhiwei Liu, Lei Zeng, Liqin Qian, Xuan Zheng, Qiong Hu, Nianzi Luo, Weinan E, Peijie Zhou, Han Wen, Yi Zhu, and Tiannan Guo. An operational perturbation proteomics-based virtual cell model. *Nature*, 2026. doi: 10.1038/s41586-026-11001-9. URL <https://doi.org/10.1038/s41586-026-11001-9>. Published online 9 September 2026.
- Jieun Sung and Wankyu Kim. scDEBART: Predicting in silico single-cell perturbation responses via large-scale differential expression learning. In *International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=pJyidZg93y>.
- Jiabin Tang, Lianghao Xia, Zhonghang Li, and Chao Huang. AI-Researcher: Autonomous scientific innovation, 2025. URL <https://arxiv.org/abs/2505.18705v1>. Version 1, 24 May 2025.
- Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, Helene Mantineo, Elizabeth M. Brydon, Zexian Zeng, X. Shirley Liu, and Patrick T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023. doi: 10.1038/s41586-023-06139-9. URL <https://doi.org/10.1038/s41586-023-06139-9>.

- Shuang Wang, Bochen Shen, Lanting Guo, Mengqi Shang, Jinze Liu, Qi Sun, and Bairong Shen. scFed: federated learning for cell type classification with scRNA-seq. *Briefings in Bioinformatics*, 25(1):bbad507, 2024. doi: 10.1093/bib/bbad507. URL <https://doi.org/10.1093/bib/bbad507>.
- Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist-v2: Workshop-level automated scientific discovery via agentic tree search, 2025. URL <https://arxiv.org/abs/2504.08066v1>. Version 1, 10 April 2025.
- Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2309.03409>.
- Jesse Zhang, Airol A. Ubas, Richard de Borja, Valentine Svensson, Nicole Thomas, Neha Thakar, Aidan Winters, Umair Khan, Matthew G. Jones, John D. Thompson, Vuong Tran, Joseph Pangallo, Efthymia Papalexi, Ajay Sapre, Hoai Nguyen, Oliver Sanderson, Maria Nigos, Olivia Kaplan, Sarah Schroeder, Bryan Hariadi, Simone Marrujo, Crina Curca Alec Salvino, Guillermo Gallareta Olivares, Ryan Koehler, Gary Geiss, Alexander Rosenberg, Charles Roco, Daniele Merico, Nima Alidoust, Hani Goodarzi, and Johnny Yu. Tahoe-100M: A giga-scale single-cell perturbation atlas for context-dependent gene function and cellular modeling. *bioRxiv*, 2025. doi: 10.1101/2025.02.20.639398. URL <https://www.biorxiv.org/content/10.1101/2025.02.20.639398v3>. Version 3, May 10, 2025.
- Zhimeng Zhou, Yang Nan, Minjie Mou, Yuntao Qian, Yixiu Liu, Zhengyu Zuo, Hao Yang, Weixian Xu, Bo Li, Wanghao Jiang, Yanlin Ren, Yang Liao, Yimeng Wang, Yinghong Li, Qingxia Yang, Zhiheng Xi, Tiantian Mi, Huaicheng Sun, Pengfei Liu, and Feng Zhu. An AI system for autonomous algorithm evolution in drug development. *bioRxiv*, 2026. doi: 10.64898/2026.08.16.745117. URL <https://doi.org/10.64898/2026.08.16.745117>. Version 1, posted 20 August 2026.

APPENDIX A. ALGORITHM AND TASK PROTOCOLS

A.1 RESEARCH ALGORITHM AND OPTIMIZATION

All three task families use the parameter update, development-selection rule and proposal interface of Section 3. A task adapter defines the input representation, prediction components, training loss and development scorer. The research process below starts from the task’s reference design. The fitted reference also supplies its first evidence card.

BioCoLoop research procedure	
1	Fit the reference design from the task initialization and seed; retain its best development checkpoint and initialize the research history.
2	Generate the shared first proposal without development feedback. From the second slot onward, supply the task, untried configurations, retained design and prior evidence cards.
3	Parse the proposal and map its <code>design_id</code> to an executable configuration. Allow at most two generation attempts in the slot.
4	Fit a valid candidate from the same initialization procedure and seed for the assigned number of aggregation rounds.
5	Select its best development checkpoint. Replace the incumbent only if its score/loss pair is better.
6	Append the executed configuration, hypothesis, diagnostics and decision to the history, including rejected or failed trials.
7	After the final slot, return the retained design, its selected checkpoint and the trial history; score that checkpoint on the held-out task.

In the controlled direct/loop comparison, the first proposal is generated once without development feedback and shared; measured history enters loop search from the second proposal. Direct search subsequently receives the default configuration and untried identifiers without development outcomes; history-guided search receives the retained configuration and accumulated cards. Both searches use the same design library, proposal count and candidate-training schedule. Because each search retains the best design visited under the same library and selection rule, the terminal retained design can coincide even when the two proposal sequences differ; Appendix B therefore reports every prespecified proposal prefix, and Section 4.4 studies how a fixed training budget is allocated from the same evidence. A configuration evaluated previously under the same task, data access and seed can reuse that completed fit; its logical position in each search remains unchanged.

Each aggregation round contains one local AdamW epoch per participating laboratory. The batch size is 64, gradient-norm clipping is one, and the local optimizer state resets each round. The learning rate and weight decay follow the selected design. The proximal penalty is zero throughout the reported design libraries. The coordinator weights local parameter states by their number of training examples. Coordinator momentum follows Equation 1 and applies only to trainable parameters. Floating-point buffers use sample-weighted averaging; integer and Boolean buffers use an elementwise maximum. These buffers store non-parameter model state. Every candidate starts with the same task-specific public or random initialization procedure and study seed. The optional residual module has width 32 and a zero-initialized output, while the original prediction head remains trainable.

The implementation calls each laboratory a *client* or *worker*; its configuration field `server_momentum` stores the coordinator momentum coefficient. These names refer to the same laboratory and coordinator roles used in Section 3.

Full-training candidates receive 100 aggregation rounds, with development evaluation every five rounds. The selection key is mean primary score followed by negative mean loss over the development panels designated by the protocol (Section 3.2). In the main experiment these panels are the participating laboratories. A complete tie retains the earlier checkpoint or incumbent. Training-example counts refer to the adapter’s training rows: for the cell tasks these are intervention-level training profiles, rather than the number of cells underlying a profile.

A.2 DESIGN LIBRARY, PROPOSALS AND EVIDENCE

The prefix D stands for *design*; its two-digit suffix is an identifier, not a training duration or ranking. A candidate is the complete executable configuration addressed by one identifier (a model specification together with its training hyperparameters), not a single generic module such as an MLP or MHA.

D00 denotes the starting configuration. The main study selects six further proposals from D01–D11 without repeating a design within a search. Each identifier maps to the complete four-factor configuration below. In this menu the only architecture change is the optional zero-initialized width-32 residual prediction module; the task adapter supplies the backbone and prediction head. The extended library and its identifiers are given in Appendix B.

Design	Learning rate	Weight decay	Momentum	Residual module
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

The main comparison has two self-improving model blocks. Qwen2.5-7B-Instruct uses bfloat16 inference, temperature 0.7, top- p 0.9 and at most 512 new tokens. For zero-indexed proposal slot t and generation attempt $b \in \{0, 1\}$, its sampling seed is $z + t + 10,000b$, where z is the study seed. GPT-5.6 Luna is queried through a text-only bridge at low reasoning effort with tools disabled and the same bounded proposal protocol; its service does not expose a generation seed. The Luna runs retain the seed labels for training and search repetition identity, but do not claim deterministic language-model sampling. An invalid identifier, a repeated configuration or a malformed response triggers the second attempt; a second failure consumes the slot without fitting a candidate. A numerically failed fit likewise leaves the incumbent unchanged and is recorded as negative evidence.

A proposal has exactly four fields:

```
{
  "hypothesis": "why the proposed change may help",
  "experiment": "the intended controlled comparison",
  "expected_effect": "the anticipated metric or diagnostic change",
  "design_id": "D08"
}
```

The three descriptive fields must be nonempty strings. The identifier must belong to the available design library. The executed configuration is obtained from that identifier; the text records the self-improving agent’s rationale, not an additional executable instruction or a measured result.

An evidence card contains the executed configuration, its hypothesis, the best development score and loss, the selected round, and the acceptance decision. Its trajectory summary includes the first and last development-evaluation rounds, their sample-weighted training losses, equal-laboratory development scores and losses, and the last-minus-first changes in development score and loss. For a 100-round fit these endpoints are rounds 5 and 100. In the main comparison, the development summary also records the worst laboratory score and the number of evaluating laboratories. Training participation and the designated evaluation panels are specified separately for source-transfer and repartitioning studies. Ordered cards are supplied as context; unsuccessful trials are not removed.

An observed proposal sequence. For Norman with ten laboratories and seed 42, D06 is accepted in the first slot: it ties D00’s development Top-1 of 40.00 but reduces development loss from 158.98 to 158.94. In the second slot, D08 retains the residual module and changes the learning rate from 10^{-3} to 10^{-4} . Its development Top-1 rises from 10.20 at round 5 to 51.90 at round 100; its development loss falls from 168.12 to 153.85. The candidate is accepted. The next proposed design, D07, reaches a best Top-1 of 45.00 and is rejected. Thus the history records both the slower-improving accepted candidate and the later unsuccessful alternative.

Three representative trajectories are summarized below. Scores and losses are multiplied by 100 and proposal slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	66.58	52.12	5 / 5
VCC, seed 43	D00	24.04	160.78	initial baseline
Norman, seed 42	D08	51.90	153.85	3 / 2

PTPC selects stronger weight decay through the loss tie-break at equal AP, whereas VCC retains its initial design. History-guided and feedback-free search share the design library, proposal count and selection rule, so they are reported together where the comparison is informative: as search efficiency in Section 4.7 and as training allocation in Section 4.4. These observations describe search progress within the twelve-design library.

A.3 INDEPENDENT TRAINING-ALLOCATION CONTROL

This study fixes the candidate set and changes only the allocation of training rounds. It uses seeds 53–55 on proteomics and the three cell settings. The ten configurations cover the following factor combinations; every configuration uses the same ten laboratories.

Learning rate	Weight decay	Momentum	Residual module
0.001	0.001	0	no
0.0001	0.001	0	no
0.0001	0.001	0	yes
0.0001	0.01	0.9	no
0.0001	0.01	0.9	yes
0.001	0.001	0.9	yes
0.01	0.001	0	yes
0.01	0.001	0.9	yes
0.01	0.01	0.9	no
0.001	0.01	0	yes

Uniform allocation trains every design for 80 rounds. Trajectory-guided allocation first trains each for 20 rounds. Within each of the three learning-rate groups, it promotes the design with the highest primary score at its best screening checkpoint, using lower development loss to break ties. From the remaining designs in that group, it also promotes the largest development-score improvement from round 5 to round 20, using lower round-20 development loss to break ties. Complete ties follow the order of the configurations in the table. The six promoted designs restart from their initial parameters and train for 100 rounds. Final selection considers their full-training checkpoints; the four unpromoted designs stop after screening.

Both schedules use 800 aggregation rounds: $10 \times 80 = 10 \times 20 + 6 \times 100$. With ten participating laboratories, this is 8,000 local epochs. Evaluation every five rounds gives 160 aggregate development evaluations, each formed from ten laboratory evaluations. These are matched training and evaluation counts; the scheduler itself generates no language-model proposals.

Across the 12 task–seed held-out comparisons, allocation gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties; VCC and PTPC tie in all three seeds. A separate TAPB development replay uses the seven-design intersection of existing trajectories: uniform allocation receives 525 candidate-rounds and trajectory-guided allocation receives 520, with a mean development AUROC difference of +0.39 points (two wins and one tie). It is not included in the twelve held-out comparisons.

A.4 DATA CONSTRUCTION AND TASK ADAPTERS

Drug–target interaction. The BindingDB partition contains 50,149 fitting and 5,604 development pairs. Canonical compounds are assigned to ten laboratories, giving 4,932–5,058 training and 517–643 development pairs per laboratory. The fixed test sets contain 5,505 random, 3,791 unseen-drug

and 2,507 unseen-protein pairs. TAPB uses frozen public protein features and trainable drug, fusion and classification components; an optional residual adapter precedes its native confounder-averaged prediction. Classification NLL and masked-language loss form its local training objective.

Observed-response proteomics. The PTPC source pool contains 333 compound groups and 491 conditions, partitioned into 386 fitting and 105 development conditions. A laboratory owns 35–42 fitting conditions; the test set contains 148 conditions. For each 6h and 24h profile, preprocessing applies $\log_{1p}$, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. The two 5,171-dimensional proteomic blocks and drug vector form a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head has protein and drug convolutional branches followed by a 64–32–1 classifier, totaling 464,193 parameters under D00. It is trained from random initialization using binary cross-entropy. All comparison methods use these same inputs and architecture; AP is the selection metric.

Cell inputs and response units. VCC, Norman and Tahoe have whole-intervention fitting/development/test splits of 60/20/20, 70/20/27 and 180/60/67. Training uses one mean response profile per intervention. VCC development and test queries average disjoint groups of up to 64 cells, omitting a final group smaller than 16 cells. Norman queries are the released scVI-derived differential-expression profiles for each gene pair. Tahoe uses one aggregate response profile per compound at the selected cell context and dose. These are the units scored by the identification task.

The scDEBART backbone receives the shared untreated reference and candidate perturbation encoding; its features are fixed while the response head is trained. The corrected feature extractor implements three attention masks as additive blocking masks. Its correction preserves the pretrained weights, gene panels, response targets, intervention partitions and candidate options. Genetic-task scoring excludes the union of directly perturbed genes. Tahoe concatenates a fixed 128-dimensional projection of a 2,048-bit Morgan fingerprint to the 768-dimensional backbone features, with the additional head-input weights initialized to zero. Its drug projection uses seed 1234. The training loss is response MSE plus 0.1 times a contrastive loss over interventions in the local training minibatch; the cosine-logit temperature is 0.1.

Five-option queries and scoring. For every development or test intervention, four distinct distractors are sampled uniformly without replacement from the *same* development or test intervention pool, excluding the true intervention. The true intervention is added and the five positions are shuffled. A single roster is fixed per intervention and reused for all of its response queries, all training seeds and all methods. Split construction and roster sampling use the preparation generator initialized with seed 20260918. Training does not use these five-option rosters; its contrastive negatives come from the local training minibatch.

For each query, predicted candidate responses and the observed response are L2-normalized over the scoring genes. Candidates are ranked by their cosine similarity; equal scores retain the fixed shuffled option order. Let G be the number of evaluated interventions, m_g the number of response queries for intervention g , and $r_{g,j}$ the rank of the true option for its j th query. The primary score is

$$\text{MacroTop1} = \frac{1}{G} \sum_{g=1}^G \frac{1}{m_g} \sum_{j=1}^{m_g} \mathbf{1}\{r_{g,j} = 1\}.$$

Thus each intervention receives equal weight even when it contributes a different number of query profiles. Micro Top-1, mean reciprocal rank and Top-3 average over individual queries. The development loss is cross-entropy over the five cosine scores divided by temperature 0.1. Development selection first computes these scores within each laboratory and then averages laboratories equally.

A.5 COMPLETE COMPARISONS AND UNCERTAINTY

The following tables give the control comparison, secondary metrics and all three drug–target test sets. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory throughout; BioCoLoop appears at one and ten laboratories. Bold marks the best displayed value within a task and metric, including ties. Higher is better except for binary cross-entropy and cell cross-entropy.

Table A.1: Control comparison for the task-reference models. The fixed and feedback-free direct references train or search on a single site, so they are reported on one laboratory; BioCoLoop is reported at one and ten laboratories so that the effect of laboratory participation can be read under the same research loop. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

Labs	Method	DTI	Proteomics	Cell perturbation		
		AUROC $\uparrow$	AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
1	Fixed model	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00
1	Direct	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
1	Our loop	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	20.22 $\pm$ 2.22	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
10	Our loop	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	22.25 $\pm$ 0.45	22.39 $\pm$ 2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

Table A.2: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44 and multiplied by 100. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	82.83	81.57	49.95	52.37
Direct	1	89.89	87.77	65.71	61.02
Our loop	1	89.89	87.77	65.71	61.02
Our loop	10	93.76	92.73	73.71	36.59
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	25.05	65.49	42.65	85.14
Direct	1	20.13	61.80	44.34	81.76
Our loop	1	20.13	61.80	44.34	81.76
Our loop	10	36.99	73.87	39.83	85.14
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	33.61	56.53	73.89	154.51
Direct	1	21.48	47.10	62.78	160.82
Our loop	1	20.74	46.94	64.26	160.86
Our loop	10	33.98	56.34	69.44	158.14
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	25.56	48.50	63.93	158.74
Direct	1	14.15	42.12	58.35	160.39
Our loop	1	14.15	42.12	58.35	160.39
Our loop	10	22.25	50.07	70.96	158.83
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	19.40	45.00	58.21	163.08
Direct	1	15.42	42.60	56.72	161.90
Our loop	1	15.42	42.60	56.72	161.90
Our loop	10	22.39	47.95	61.19	160.75

Table A.3: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Our loop	10	93.76	92.73
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Our loop	10	91.65	92.58
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Our loop	10	87.63	85.07

Three-seed means and sample standard deviations summarize training/search variability on fixed partitions. Sample-level intervals use paired cluster bootstrap with 1,000 fixed-seed resamples of

retained predictions. Drug identity defines clusters for the random and unseen-drug interaction sets, and protein sequence defines clusters for the unseen-protein set. Proteomic and cell intervals use compound and intervention clusters, respectively. Both methods in a contrast receive the same re-sampled clusters. The training-allocation study instead uses 5,000 hierarchical paired resamples: training seeds are drawn first, followed by compound or intervention groups within each sampled seed.

The accompanying *BioCoLoop Statistical Supplement*, Tables S.1–S.2, reports the complete per-seed differences and confidence intervals for the main contrasts and all three drug–target test sets. It includes every reported contrast and seed, including zero and negative differences. These model-conditional intervals quantify test-sample uncertainty; in several cell comparisons they include zero, so positive mean differences do not uniformly establish significant gains.

The accompanying research records contain the executable configurations, proposals, development histories and saved predictions needed to reconstruct these comparisons. The reproducibility statement describes their availability and verification.

A.6 SCOPE AND LIMITATIONS

All experiments simulate separate laboratories on a single host and use disjoint partitions of public data, so they do not exercise real non-IID device populations, communication cost, stragglers or secure aggregation, and no differential-privacy or update-inversion analysis is provided; the framework should be read as a research interface, not a deployable privacy mechanism. The design library covers four training factors with one optional residual module, so model development here means selecting a model specification and its training hyperparameters from a registered menu rather than architecture search. Half of the main comparison uses a proprietary service without a reproducible generation seed, and the public self-improving agents are task-adapted re-implementations that do not complete every endpoint. The identification tasks have a 20% chance level and small headroom: every arm lies in 7.5–31.6% macro Top-1 against it and between 36.00 and 57.00 MRR against a 46.00 chance level, and the untrained public head is itself at 7.5% on Norman, so these values reflect the frozen scDEBART representation rather than the scoring path. The rosters themselves are sound: the true option is placed near-uniformly across them and a prototype oracle built from observed held-out responses retrieves it at 94.5% (VCC) and 100% (Norman). Measured history changes how quickly the search reaches a good design, so we report it as search efficiency in Section 4.7 and as training allocation in Section 4.4. A random sweep of the registered library is the natural additional control for design search, alongside the feedback-free direct search used in Appendix A. Additional controls are discussed below.

Data access is the object of study rather than an uncontrolled advantage. The single-laboratory references use laboratory 0 because that is the setting their fixed or centralized protocols support, while BioCoLoop uses the ten-laboratory pool that collaborative learning is defined to exploit, and the control of Table A.1 varies laboratory participation under the same research loop to separate the two. This design still does not compare collaboration against a centralized pool of the same ten datasets: a pooled-data upper reference would quantify how much of the benefit comes from data volume rather than from distributional breadth, and our source-as-laboratory studies already show that additional sources help only when their measurement context is compatible. Reporting a ten-laboratory arm is therefore a deliberate statement about the regime we study, not an attempt to win comparisons against single-site baselines.

Further scope limits the claims. The two evidence-driven policies are evaluated separately, on a small registered design menu and three seeds per endpoint, and they are never combined into a jointly tuned algorithm; the reported intervals are conditional on the retained models, and several cell-endpoint intervals include zero. Training and evaluation are simulated on one host with one shared implementation, so the study does not price communication, synchronization or straggler effects, and the residual-module and random-sweep comparisons cover only a narrow part of the design space. The proposal-history policy is also a fixed-budget search mechanism: its measured effect is how quickly a good design is reached and how a fixed training budget is reallocated, which need not extend to much larger libraries, much longer horizons, or research objectives whose development and held-out quantities diverge.

APPENDIX B. LABORATORY, SOURCE AND PROPOSER SENSITIVITY

Sensitivity study design. We examine collaboration scale, source diversity, proposer choice and research budget using the same target scorers as the main study. Laboratory-count experiments cover all five endpoints at $K \in \{1, 2, 5, 10\}$ under both increasing participation and a fixed total data pool. DTI uses seed 61; the lighter proteomics and cell experiments use seeds 61–63. Independent-source DTI and proteomics use seed 61, and external-source VCC uses seeds 61–63. The two-proposer study has two separate protocols: a 12-design, six-slot study on all five endpoints (DTI seed 42; other endpoints seed 61), and a 36-design, 24-slot study on the four lighter endpoints at seed 61. Each candidate receives 100 training rounds. The latter changes the design space as well as the search budget, so budget comparisons are made within a protocol. Main-table results retain their original seeds 42–44 and protocol.

B.1 NUMBER OF LABORATORIES

The participation experiment uses a fixed nested set of the first K original laboratories, increasing both training and development access. The fixed-pool experiment merges the complete ten-laboratory pool into K training groups while retaining equal-weight evaluation over the original ten development panels. Thus its selection population stays fixed as training partitions change. The identical $K = 10$ fit is reused by both comparisons. Every unique fit uses the default design and 100 rounds. The cell objective forms contrastive negatives within each local training group, so repartitioning changes the negative pool as well as optimization steps. These comparisons characterize the implemented training pipeline at different collaboration scales. Zero between-seed SD can occur when a fixed pretrained head receives an identical single-batch update; it describes training variability on these fixed partitions.

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing K also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	83.65	84.43	89.62	93.88
PTPC	AP	26.03 $\pm$ 5.57	37.38 $\pm$ 0.60	29.63 $\pm$ 2.50	37.54 $\pm$ 0.27
VCC	Macro Top-1	30.02 $\pm$ 0.78	39.56 $\pm$ 0.11	31.88 $\pm$ 0.00	31.48 $\pm$ 0.16
Norman	Macro Top-1	25.56 $\pm$ 0.00	18.22 $\pm$ 0.00	19.68 $\pm$ 1.11	15.48 $\pm$ 0.00
Tahoe	Macro Top-1	19.40 $\pm$ 0.00	25.37 $\pm$ 0.00	20.90 $\pm$ 0.00	26.37 $\pm$ 0.86

Table B.2: Fixed data pool. All target data are retained and merged into K clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	85.47	88.96	88.05	93.88
PTPC	AP	34.59 $\pm$ 5.66	37.36 $\pm$ 1.89	39.19 $\pm$ 0.43	37.54 $\pm$ 0.27
VCC	Macro Top-1	29.49 $\pm$ 0.00	28.80 $\pm$ 0.20	31.06 $\pm$ 0.22	31.48 $\pm$ 0.16
Norman	Macro Top-1	49.63 $\pm$ 4.20	19.56 $\pm$ 0.20	26.12 $\pm$ 0.11	15.48 $\pm$ 0.00
Tahoe	Macro Top-1	22.39 $\pm$ 0.00	21.39 $\pm$ 0.86	20.40 $\pm$ 0.86	26.37 $\pm$ 0.86

B.2 INDEPENDENT SOURCES AND TARGET EVALUATION

For DTI and cell transfer, the entire target training pool forms one laboratory, and each auxiliary collection contributes a separate local update through sample-weighted aggregation. Table B.3 distinguishes these source-defined laboratories from the within-corpus partitions used in the laboratory-count study. Only the original target development panels select the checkpoint; every configuration is evaluated on the same target test set. Target-only, individual-source and combined-source configurations start from the same task initialization.

DTI auxiliaries are BioSNAP, Davis and Human from the pinned TAPB release, with 19,092, 1,721 and 2,929 retained training pairs after drug–protein-pair de-duplication against target splits and pre-

Table B.3: Source and context mapping in the cross-source experiments. The DTI units are database collections, the cell units preserve their study and biological scenario, and each proteomic auxiliary study owns one client and its response head. The ten PTPC clients remain partitions of one target corpus. Biological contexts within a study are not counted as additional independent studies.

Task	Client unit	Sources and biological contexts
DTI	Four data collections	Target BindingDB; BioSNAP; Davis; Human. Each collection contributes one client after pair-level de-duplication.
Cell	Four study/scenario clients	Target VCC; Replogle K562; Nadig HepG2; Jiang IFNB/K562. Each auxiliary source contributes one client.
Proteomics	Ten target partitions plus three source clients	Clients 0–9: PTPC partitions; client 10: Lin (A549, H292, PC9, PSC1); client 11: Ruprecht (A549, Calu6, Calu1, 2030, 2122); client 12: decryptE (Jurkat, five doses).

ceding sources. All five DTI source configurations complete the same 100-round schedule. VCC auxiliary data comprise Replogle K562, Nadig HepG2 and Jiang IFNB/K562, with respectively 60, 60 and 58 retained training interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source feature panels preserve measured genes, and target development/test intervention identities are excluded from source training. The source manifest records retained subsets, feature intersections and exclusions.

Table B.4: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean $\pm$ SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. The two-source PTPC protocol shown here uses Lin and Ruprecht; the three-source extension is reported separately in Table B.7. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

Target	Additional source	Metric	Score $\uparrow$	Δ (pp)
DTI	Target only	AUROC	85.47	+0.00
DTI	+ BioSNAP	AUROC	82.44	-3.03
DTI	+ Davis	AUROC	87.86	+2.39
DTI	+ Human	AUROC	87.95	+2.48
DTI	+ All three	AUROC	84.30	-1.18
VCC	Target only	Macro Top-1	29.49 $\pm$ 0.00	+0.00 $\pm$ 0.00
VCC	+ Replogle	Macro Top-1	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08
VCC	+ Nadig	Macro Top-1	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18
VCC	+ Jiang	Macro Top-1	33.07 $\pm$ 0.00	+3.58 $\pm$ 0.00
VCC	+ All three	Macro Top-1	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00
PTPC auxiliary	Target only	AP	34.98	+0.00
PTPC auxiliary	+ Lin	AP	35.01	+0.03
PTPC auxiliary	+ Ruprecht	AP	35.50	+0.52
PTPC auxiliary	+ Both	AP	35.31	+0.34

The cell study reports mean and SD across three seeds together with a paired intervention-bootstrap analysis. Jiang improves target Top-1 by 3.58 points; its 95% bootstrap interval is $[-2.11, 9.51]$. This positive point estimate motivates further source-aware evaluation, while the complete table also exposes sources whose updates reduce the target metric.

Matched source–scenario laboratories. Table B.5 compares four fixed-design settings: 15 VCC training interventions in one laboratory, all 60 VCC interventions in one laboratory, the same 60 partitioned into four laboratories, and four source-defined laboratories with 15 interventions each. The latter uses VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562. Both four-laboratory arms share the same 15-intervention VCC anchor. Hash-based sampling fixes the auxiliary identities before fitting, excluding target training, development and test intervention identities as well as duplicates across auxiliary laboratories. This experiment matches training intervention counts, not raw-cell counts: the same-source arm represents 77,771 cells, versus 33,469 in the cross-scenario arm. Each source retains its measured gene panel and corresponding loss mask.

Table B.5: Fixed-design source–scenario ablation on VCC. K : training clients; N : training intervention conditions. The two $K = 4$, $N = 60$ arms share a 15-condition VCC anchor and differ in the sources of the other three clients. All arms use 100 rounds and the same target evaluation. Mean $\pm$ sample SD over three seeds; best means are bold. Cell counts and gene panels remain source-specific; protocol details are in Appendix B.

Training sources	K	N	Top-1 (%) $\uparrow$	MRR (%) $\uparrow$
VCC, limited data	1	15	29.36 $\pm$ 0.26	56.97 $\pm$ 0.19
VCC, all target data	1	60	29.43 $\pm$ 0.09	56.00 $\pm$ 0.09
VCC, same-source split	4	60	30.70 $\pm$ 0.42	56.24 $\pm$ 0.04
VCC + external studies	4	60	28.82 $\pm$ 0.00	56.30 $\pm$ 0.00

All four settings use identical head initialization, the default design, 100 rounds and seeds 61–63. The original ten target-development panels select checkpoints; the common target test contains 20 interventions and 360 five-option queries. No proposal search is performed. All twelve fits complete, and saved predictions reproduce the scorer’s metrics under independent evaluation. Cross-scenario Top-1 is lower in each seed, with a mean difference of -1.88 points relative to the same-source four-laboratory setting. The near-zero MRR difference shows that source composition can affect the top-ranked choice differently from the full ranking. The protocol and source identities remain fixed throughout the comparison.

Two-source proteomic transfer and proposal search. Two independent lung-cancer studies provide auxiliary measurements: Lin et al. (Lin et al., 2026) and Ruprecht et al. (Ruprecht et al., 2020). Their continuous response labels differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed source-specific prediction heads. Target fitting uses its binary efficacy objective; Lin predicts continuous CellTiter-Glo response, and Ruprecht predicts an interval-valued potency outcome. Target data retain the original ten laboratories; Lin and Ruprecht each add one study-level laboratory. Shared encoders use sample-weighted aggregation, whereas each private head is updated only by its own source laboratories. The common laboratory and coordinator interfaces are reused. This adapter is compared against its own target-only control, separately from the main-table PTPC predictor.

The Lin subset contains 28 drug conditions (20 source-training, eight source-development), spanning five training and two development drugs. Inputs are 24-hour, 10- μ M protein log-fold changes, and the response is the mean of the three released CellTiter-Glo ratio measurements. The HCT16/HCT116 alias is excluded from auxiliary training. The Ruprecht subset contains 109 matched drug conditions (85 training, 24 development; 17 and five drugs respectively). Protein log-fold changes at 24 hours are paired with CellTiter-Glo EC50 values in $\log_{10}$ nM: 49 exact, 18 approximate and 42 right-censored outcomes. Squared distance to the encoded potency interval is the loss. The adapter treats explicit lower bounds and values at or above 10,000 nM as right-censored and encodes approximate values with a factor-of-two interval; these are analysis conventions. Profiling dose is not a model input; source acquisition nevertheless uses dose conditions chosen in relation to drug response. The target retains 386 training and 105 development examples, with 148 held-out observations across 92 compounds and 22 positives.

All four source settings receive the same 100-round procedure. A two-proposal Luna loop is also executed independently for each setting. These 12 fits and eight valid proposal calls produce eight fixed/loop evaluations. Fixed and loop selection have identical held-out ranking scores in each source arm (Table B.6). The source effect and the proposal-history effect are thus measured separately.

Target-only AP/AUROC are 34.98/74.75. Adding Lin yields 35.01/74.78, Ruprecht 35.50/74.28 and both 35.31/74.17. Paired compound bootstrap (1,000 resamples) gives 95% intervals for the AP changes of $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points, respectively. These results measure transfer from two independent external studies and provide the source-matched controls for the two-proposal comparison.

Three-source proteomic transfer. The third auxiliary study, decryptE (Eckert et al., 2025), measures Jurkat proteomes after 18 hours of drug exposure at 1, 10, 100, 1,000 and 10,000 nM. Its source objective predicts the released continuous metabolic-activity measurement relative to DMSO. We align compounds using the published names, explicit alternative names and structures, and match

Table B.6: Independent-source proteomics auxiliary pilot, seed 61. Fixed design and a two-proposal Luna loop use the same shared-encoder/source-specific-head model and 100-round training allocation. Every source arm and both selection modes are retained. The heldout AP gain from this short loop is zero in all four source conditions; source-transfer gains are a different comparison.

Training sources	Selection	AP $\uparrow$	Δ vs target only (pp)
Target only	Fixed	34.98	+0.00
Target only	Loop	34.98	+0.00
+ Lin	Fixed	35.01	+0.03
+ Lin	Loop	35.01	+0.03
+ Ruprecht	Fixed	35.50	+0.52
+ Ruprecht	Loop	35.50	+0.52
+ Both	Fixed	35.31	+0.34
+ Both	Loop	35.31	+0.34

protein features by UniProt identifier. Inputs comprise log-transformed measured normalized protein intensities, an observation mask, drug fingerprints and dose indicators. Splits group all doses of the same chemical graph together; normalization is fitted on source-training measurements. Target development and test compounds are excluded by canonical identity and stereo-agnostic chemical graph.

The repository’s 145 main processed matrices contain one additional manually classified Vincristine copy; the uniform standard release therefore provides 144 compounds. Pre-training identity checks exclude 12 compounds overlapping the target development/test sets, six records with unsupported internal names and two records with conflicting released chemical-identity/structure annotations. The resulting source contains 124 distinct compounds and 620 dose conditions: 495 training observations from 99 compounds and 125 development observations from 25 compounds. The source manifest records every exclusion and preserves the original annotations.

We append a private continuous-response head to the shared adapter and instantiate this same four-head model in every control. All original encoder and head tensors retain identical seed-61 initialization. The four prespecified arms are target only, target plus decryptE, target plus Lin and Ruprecht, and target plus all three studies. Every arm completes 100 rounds with the same local optimizer and sample-weighted shared-parameter aggregation. Checkpoint selection uses equal-laboratory target-development AP with binary cross-entropy as the tiebreaker; auxiliary development scores are diagnostic. These fixed-design comparisons use zero proposal slots and are separate from the preceding two-source research-loop study.

Table B.7: Three-source proteomic transfer on PTPC. All four arms use the same four-head adapter, seed 61, 100-round fitting schedule and target-development checkpoint rule. AP and AUROC are multiplied by 100; bold identifies the best value in each metric. The target test set contains 148 observations from 92 compounds. These are fixed-design source comparisons; the separate two-source proposal study appears in Table B.6.

Additional source	AP $\uparrow$	AUROC $\uparrow$	Δ AP (pp)
Target only	34.98	74.75	+0.00
+ decryptE	35.03	74.21	+0.05
+ Lin + Ruprecht	35.31	74.17	+0.34
+ Lin + Ruprecht + decryptE	35.68	74.28	+0.70

Combining all three sources raises target AP from 34.98 to 35.68, a 0.70-point gain. Relative to Lin and Ruprecht together, adding decryptE increases AP by 0.37 points and AUROC by 0.11 points. Target-only training retains the highest AUROC across the four arms (74.75 versus 74.28 for all three), showing that the primary AP improvement does not imply uniform gains across ranking metrics. These source-transfer point estimates come from one seed. The four complete CPU fits take 101.21 seconds in total; saved checkpoints and predictions are sealed and independently rescored with exact agreement.

One study per training laboratory. To assign every training laboratory to a distinct source, we additionally merge all 386 target training examples into one laboratory. Lin, Ruprecht and decryptE each remain a separate source laboratory. The four arms use $K = 1, 2, 3, 4$ laboratories and respectively 386, 881, 491 and 986 training conditions. Their four-head architecture, seed-61 initialization,

Table B.8: One worker per proteomic study. The target training pool is merged into one client; Lin, Ruprecht and decryptE each form one additional client. K counts training clients and N counts training conditions. All four fixed-design arms use 100 rounds, the same initialization (seed 61), and equal-weight evaluation over the original ten target development panels. AP and AUROC are percentages on the common target held-out set; best values are bold.

Training sources	K	N	AP $\uparrow$	AUROC $\uparrow$
Target study	1	386	34.41	72.33
+ decryptE	2	881	36.58	73.81
+ Lin + Ruprecht	3	491	36.09	72.69
+ Lin + Ruprecht + decryptE	4	986	37.53	74.39

local loss, sample-weighted shared-parameter aggregation and source-private head ownership are unchanged. The merged target laboratory performs seven minibatches per local epoch, instead of aggregating updates from ten smaller target partitions. This is therefore a separately trained protocol, with a matching target-only control.

The original ten target-development panels still select checkpoints with equal panel weights, independently of the number of training laboratories. All four configurations complete 100 rounds and use zero proposal slots. Held-out AP and AUROC are independently recomputed from saved predictions, with maximum absolute discrepancy 1.11×10^{-16} . The full CPU run takes 76.18 seconds.

With one laboratory per study, all three auxiliary sources improve target AP from 34.41 to 37.53 and AUROC from 72.33 to 74.39. Adding decryptE to Lin and Ruprecht increases AP by 1.44 points. These single-seed source-transfer comparisons vary available training data as well as participating studies; they measure the effect of those sources under a fixed design, separately from proposal feedback and the matched-condition cell study.

Within-study proteomic contexts. HCC1143, HCC1806 and MDA-MB-453-1 are drawn from the mtPTDS supplementary collection (Sun et al., 2026), each with 47 retained drug conditions and 3,501 proteins shared with the target panel. They are three biological contexts within one study. Target development/test compounds are excluded from source training. These experiments use the original proteomic adapter and seeds 61–63.

Table B.9: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean $\pm$ sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

Additional context	AP $\uparrow$	Δ AP (pp)	Paired 95% CI
Target only	34.59 $\pm$ 5.66	+0.00 $\pm$ 0.00	[0.00, 0.00]
+ HCC1143	34.94 $\pm$ 5.15	+0.35 $\pm$ 0.56	[-1.22, 1.60]
+ HCC1806	35.23 $\pm$ 5.35	+0.64 $\pm$ 0.32	[-0.78, 1.77]
+ MDA-MB-453-1	35.20 $\pm$ 5.31	+0.61 $\pm$ 0.44	[-0.85, 2.04]
+ All three	35.47 $\pm$ 6.36	+0.87 $\pm$ 8.89	[-2.44, 3.99]

B.3 RESEARCH-MODEL AND SIX-SLOT PROPOSAL COMPARISON

The short study uses the 12 executable designs of Appendix A. Direct search receives untried identifiers without measured scores; the loop additionally receives accumulated development evidence. Both modes share initialization and the first proposal. Each slot allows two attempts to return the four-field proposal schema; failed slots consume budget and remain in the trace. Every valid candidate restarts from the common task/seed initialization and is fitted for 100 rounds. Prefixes at 0, 1, 2, 4 and 6 slots are evaluated after the trajectory is complete. An exact fit cache can reuse an identical configuration while retaining its logical training-budget charge.

Qwen uses the local sampling settings in Appendix A. The second proposer is the gpt-5.6-luna service, accessed in a text-only session with tool use disabled; request metadata and responses are retained. The comparison holds task messages, editable designs, training procedure and proposal

budget fixed across backends. Service-level decoding controls and parameter count are not exposed by the Luna interface.

Table B.10: Original 12-design/six-slot heldout results. DTI seed 42; four light endpoints seed 61. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Fixed	Direct@6	Loop@6
DTI	Luna	94.27	94.60	94.27
DTI	Qwen	94.27	94.27	94.27
PTPC	Luna	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27
VCC	Luna	31.50	31.50	31.50
VCC	Qwen	31.50	31.50	31.50
Norman	Luna	15.48	22.30	23.11
Norman	Qwen	15.48	22.30	22.30
Tahoe	Luna	25.37	26.87	26.87
Tahoe	Qwen	25.37	25.37	25.37

Table B.11: Original 12-design/six-slot search dynamics. DTI seed 42; four light endpoints seed 61. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
DTI	Luna	94.71	0/0	6/0	6/0	6/6
DTI	Qwen	94.71	0/0	0/0	0/0	6/6
PTPC	Luna	66.72	0/0	0/0	0/0	6/6
PTPC	Qwen	66.72	0/0	0/0	0/0	6/5
VCC	Luna	24.04	3/6	3/6	3/6	6/6
VCC	Qwen	23.65	0/0	0/0	0/0	6/6
Norman	Luna	51.70	5/5	5/5	5/5	6/6
Norman	Qwen	51.70	3/4	3/4	3/4	6/6
Tahoe	Luna	30.00	0/0	0/0	6/3	6/6
Tahoe	Qwen	30.00	0/0	0/0	0/0	4/4

All five endpoints have both backends and both search modes. Among the ten direct/loop pairs, Norman/Luna improves by 0.81 Top-1 points, DTI/Luna decreases by 0.33 AUROC points and eight pairs tie. The twenty DTI prefix scores were independently recomputed from saved predictions for all 5,505 test pairs, with checkpoint and output hashes verified. Figure 4 shows the complete prespecified prefix sequence for every endpoint, alongside its development selection trajectory.

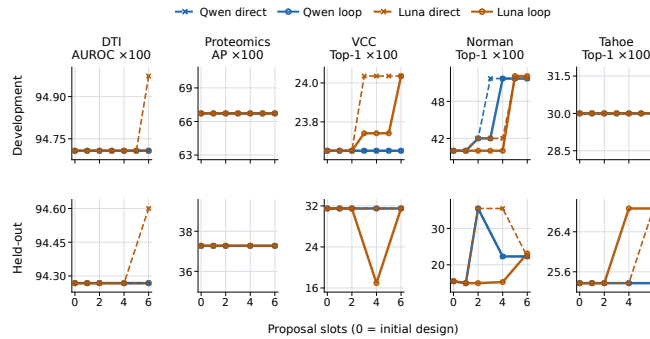


Figure 4: Research trajectories with and without evidence history in the six-slot protocol. Curves report development selection and held-out scores at each proposal prefix; each candidate receives 100 rounds and slot 0 is the initial design.

B.4 EXTENDED SEARCH AND TASK-DEPENDENT SATURATION

The extended study increases the design library to 36 configurations: three learning rates, three weight decays, two coordinator-momentum values and two residual-head settings. It runs 24 proposal slots for both backends and modes on PTPC, VCC, Norman and Tahoe. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. The settings for initialization, full candidate fitting, development selection and invalid-slot accounting are otherwise unchanged. DTI’s backend comparison is the six-slot study in B.3; it is not part of this extended protocol.

Table B.12: Extended 36-design/24-slot heldout results. Four light endpoints, seed 61; no native DTI 24-slot result is implied. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Fixed	Direct@6	Loop@6	Direct@24	Loop@24
PTPC	Luna	37.27	37.27	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27	37.27	37.27
VCC	Luna	31.50	16.94	16.94	20.23	20.01
VCC	Qwen	31.50	31.50	31.50	19.61	19.61
Norman	Luna	15.48	35.48	23.11	23.11	23.11
Norman	Qwen	15.48	22.30	22.30	22.30	22.30
Tahoe	Luna	25.37	13.43	26.87	13.43	13.43
Tahoe	Qwen	25.37	26.87	26.87	13.43	13.43

Table B.13: Extended 36-design/24-slot search dynamics. Four light endpoints, seed 61; no native DTI 24-slot result is implied. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
PTPC	Luna	66.72	0/0	0/0	0/0	24/24
PTPC	Qwen	66.72	0/0	0/0	0/0	19/14
VCC	Luna	27.68	16/9	16/10	16/10	24/24
VCC	Qwen	24.83	18/23	18/23	18/23	11/16
Norman	Luna	52.10	22/4	22/4	22/7	24/24
Norman	Qwen	51.70	2/5	2/5	22/19	15/12
Tahoe	Luna	35.00	5/13	5/13	5/13	24/24
Tahoe	Qwen	35.00	18/14	18/14	18/14	14/15

Luna produces valid designs in all 192 attempted slots, and Qwen in 116 of 192. Across 16 mode trajectories, the primary development score is unchanged in the final eight slots in 12 cases; the full score/loss selection key is unchanged in ten. Each trajectory retains 11–24 unvisited designs. A plateau therefore characterizes finite-budget search with a particular backend, rather than exhaustion of the design space.

Table B.13 measures the first slot attaining the smaller terminal development score within each direct/loop pair, a common target reached by both modes. Luna reaches Norman’s common target earlier with feedback (4 versus 22 slots), whereas direct search reaches Tahoe’s target earlier (5 versus 13). Proteomics retains the initial design. Table B.15 connects Norman’s trajectory to concrete configuration changes: reducing the learning rate to 10^{-4} gives an early development gain, and removing the residual head reaches the final primary-score target. Subsequent loss-tiebreak selections can change the retained design without increasing that primary score.

Within this extended protocol, Qwen’s VCC held-out Top-1 decreases from 31.50 at slot 6 to 19.61 at slot 24 while development selection improves. This separates the speed of finding a strong development design from generalization of the selected predictor. The complete development and held-out curves support task-specific budget analysis alongside the proposal-validity and remaining-space diagnostics.

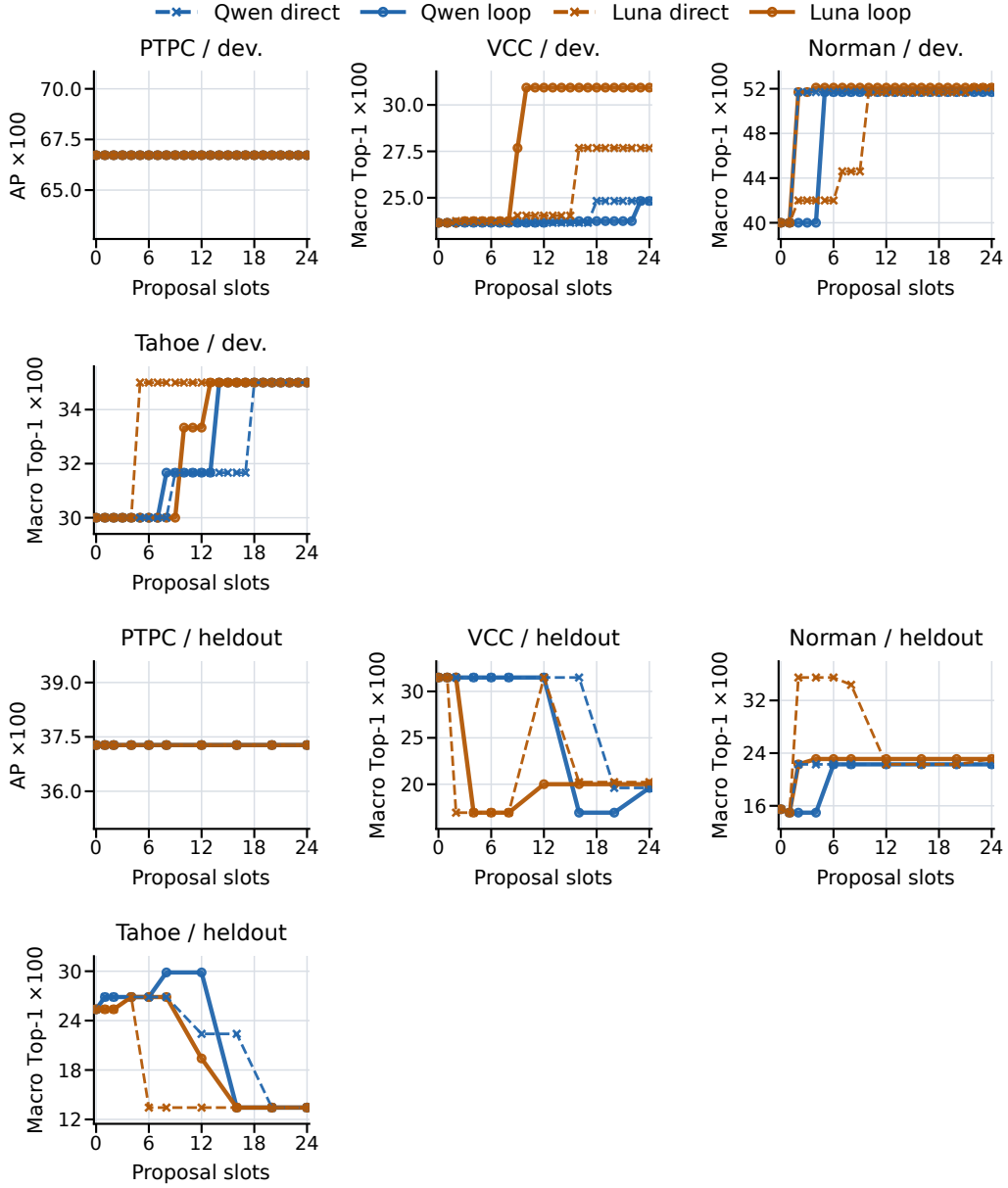


Figure 5: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

B.5 EXECUTED DESIGNS AND PROPOSAL TRACES

The short protocol uses exactly the D00–D11 configurations listed in Appendix A.2. The extended protocol uses all $3 \times 3 \times 2 \times 2 = 36$ combinations in Table B.14. An identifier $L_iW_jM_kR_l$ selects the indexed value of each factor, with $i, j \in \{0, 1, 2\}$ and $k, l \in \{0, 1\}$. For example, LOW1M0R1 specifies learning rate 10^{-4} , weight decay 10^{-3} , zero coordinator momentum and an enabled residual module. Every configuration has proximal coefficient zero; the task-specific prediction interface remains fixed.

Table B.14: Exact factor-to-index mapping for the extended 36-design library. Identifiers concatenate the four prefix–index pairs in L, W, M, R order; all combinations of the listed values are included.

Factor	Identifier prefix	Index 0	Index 1	Index 2
Learning rate	L	10^{-4}	10^{-3}	10^{-2}
Weight decay	W	0	10^{-3}	10^{-2}
Coordinator momentum	M	0	0.9	–
Residual module	R	disabled	enabled	–

The accompanying reproducibility archive retains all 504 attempted proposal slots and their acceptance decisions: 120 slots in the short protocol and 384 in the extended protocol. Its machine-readable proposal trace records each proposal’s full configuration, changes from the incumbent and measured development score; the expanded trace tables are provided with the archive rather than reproduced in the article. Together with the prefix curves, these records connect research decisions to both the executable intervention and its observed outcome.

Table B.15: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

Mode	Slot	Executed configuration	Dev. Top-1
Direct	0	Common initial recipe	40.00
Direct	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Direct	2	lr=0.001, wd=0.001, m=0.9, R=1	42.00
Direct	7	lr=0.001, wd=0.01, m=0.9, R=1	44.60
Direct	10	lr=0.0001, wd=0, m=0, R=1	51.70
Direct	11	lr=0.0001, wd=0.001, m=0, R=1	51.70
Direct	22	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	0	Common initial recipe	40.00
Loop	1	lr=0.001, wd=0.001, m=0, R=1	40.00
Loop	2	lr=0.0001, wd=0.001, m=0, R=1	51.70
Loop	4	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	7	lr=0.0001, wd=0, m=0, R=0	52.10

APPENDIX C. PUBLIC SELF-IMPROVING-AGENT COMPARISON

C.1 SHARED BIOLOGICAL TASK INTERFACE

We connect the released self-improving agents of AI-Scientist-v2 (Yamada et al., 2025) and AI-Researcher (Tang et al., 2025) to the same biological fitting and evaluation service used by BioCoLoop. The source versions are 96bd51617cfd and f9a6f8480860, respectively (SakanaAI, 2025; HKUDS, 2025). Because neither public framework implements our collaborative training and distributed-evaluation protocol, each public self-improving agent accesses laboratory 0 only; BioCoLoop accesses all ten laboratories. The agents otherwise share the original twelve-design library, at most six candidate proposals, 100 training rounds per valid candidate and the same held-out scorer. The service first fits D00 afresh. The submitted design ID is the executable field; its accompanying hypothesis, experiment and expected-effect text records the controller’s rationale. Repeated or invalid design requests consume a proposal slot, and an agent may stop its native refinement process before exhausting the cap. Development score and loss select the retained checkpoint under the original selection rule.

AI-Scientist-v2 retains its experiment manager, tree-search paths, experiment reviews and journal. AI-Researcher retains its native idea selection, planning, implementation, review handoffs and iterative experiment analysis. Task-specific prompts direct these agents to the registered design interface. Executing arbitrary generated code, external retrieval, figure review and manuscript generation are outside this fixed-task experiment. We run two matched controller blocks: local Qwen2.5-7B-Instruct with temperature 0.7, top- p 0.9 and at most 512 generated tokens per call, and GPT-5.6 Luna at low reasoning effort through a text-only, tool-disabled bridge. Their adapters allow at most 160 model calls and 28,000 input tokens per call, raised to 65,536 for the four starred GPT-5.6 Luna AI-Researcher retries described in Section C.2. Qwen uses registered generation seeds; the Luna service exposes no generation seed, so seed labels identify training/search repetitions only. Full model-call counts, token usage and transport receipts accompany each run; equal proposal and candidate-training budgets do not imply equal language-model costs.

C.2 PREDICTION QUALITY AND EXECUTION OUTCOMES

A scored run completes the research process selected by its self-improving agent under the registered cap, seals development selection and passes independent rescoring of saved held-out predictions. Table 1 reports matched Qwen2.5-7B-Instruct and GPT-5.6 Luna blocks summarized over seeds 42–44. Fixed, direct and BioCoLoop rows are complete three-seed summaries; a cell of a public self-improving agent that terminates before a valid held-out selection keeps that outcome instead of an imputed score, marked in the table with a dagger when it is partial and left as an unscored dash otherwise. The partial cells are Qwen2.5-7B-Instruct AI-Researcher on VCC (2/3 seeds, F_{tool}) and GPT-5.6 Luna AI-Scientist-v2 on Tahoe (2/3, F_{svc}).

Four GPT-5.6 Luna AI-Researcher cells were first registered as unscored because the agent’s accumulated native history exceeded the 28,000-token per-call admission (F_{ctx}). That limit is transport bookkeeping rather than part of the research task, so those cells were re-run with the same tasks, seeds, design library, six-proposal cap, candidate training and held-out scorer, admitting at most 65,536 input tokens per call; the four completed cells carry a star in the table and their registered per-seed records are retained. The BindingDB (DTI) cell remains unscored (0/3): its native refinement trajectory is the longest of the five endpoints, one frozen seed reached 29,167 input tokens and was refused at the registered admission (F_{ctx}), the other two terminated in the Luna service transport (F_{svc}), and the registered retries under the extended admission likewise ended in that transport without sealing a held-out selection. That outcome is reported rather than repaired by further controller changes. The grouped table therefore makes research-model identity, access, self-improving-agent identity and execution coverage visible in one primary comparison, and the final column summarizes each method’s mean rank across the endpoints it completed.

Failure tokens are typed rather than collapsed into missing values. F_{pipe} denotes a native pipeline termination in a public self-improving agent before selection is sealed, F_{tool} a native tool-transport termination, F_{ctx} a registered context-limit termination and F_{svc} a Luna service-transport termination. These outcomes remain in the per-seed receipts and are recorded above for each incomplete table cell.

The seed-42 Qwen AI-Scientist-v2 DTI example trains three candidates before its native substage generator fails schema validation, so the unsealed incumbent is not scored.

The held-out scorer runs only after development selection is sealed. For every scored cell, an independent verifier reloads the saved predictions, recomputes the task metric and checks the selected checkpoint and artifact hashes. The DTI rescore differs from the stored metrics by at most 10^{-15} . Raw model calls, native stage history, candidate requests, selections and verification receipts accompany the runs.

C.3 TASK-TRANSPORT REPAIR

The final AI-Researcher adaptation removes repeated task packets from stage prompts while retaining the complete current task and measured evidence in each system prompt. Native stage reasoning and tool results are preserved. The text bridge accepts an unambiguous `parameters` alias for `arguments` and standard single-tool-call envelopes, without inferring actions or parameter values. Six proposals are a maximum, so the self-improving agent may end refinement early. Model weights, context allowance, candidate training and scoring are unchanged. This transport revision is fixed across the five reported endpoints; the Luna sensitivity uses the same interface. Failures are retained as execution outcomes rather than motivating post-hoc controller repairs.